

RIOT: Your Friend For The IoT

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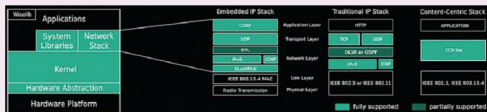
As a proud father would speak of his daughter, the developers of RIOT state, “RIOT powers the Internet of Things (IoT) like Linux powers the Internet.” A free software licensed by GNU lesser general-public licence (LGPL)-v2, RIOT is an operating system (OS) developed by a grassroots community with help from companies, academia and hobbyists worldwide.

The moment you begin working on a project for the IoT, you are pulled into tight waters, having to provide maximum flexibility at minimum resource usage. Low memory footprint and high energy efficiency are the most basic requirements, along with support for real-time functioning. RIOT is structured keeping all these factors in mind, making it perfectly suited for IoT applications.

A micro-kernel that holds up the OS in its stead

Usage of micro-kernel architecture in RIOT OS allows for a simple abstraction layer over the hardware and provides the necessary mechanisms for the OS to function right. Well-designed thread management and a priority based scheduler allow for the smooth processing of tasks. This OS exploits ultra-low interrupt latency and priority based scheduling to provide real-time action.

The micro-kernel includes a powerful application program interface (API) that takes care of inter-process communication. Mutexes are engaged to synchronise the performance between tasks, with a system timer maintaining the schedule perfectly.



RIOT's network architecture

What RIOT features

Built with the sole purpose of developing applications for the IoT, RIOT implements all relevant open IoT standards and works at keeping it connected, secure and long lasting. Currently, it is partly compliant with portable OS interface (POSIX) and designed such that if you code once for an 8-bit platform, you can easily port it to 16- and 32-bit platforms as well. Claimed to be developer-friendly, resource-friendly and IoT-friendly, this article explores what it means to develop with RIOT.

Emphasising low-power operation at every stage

Once the board that is to run the OS is initialised, it goes into the lowest possible power mode, running the idle thread. Once the main thread is given priority, the board switches threads to perform the task called by the main function.

Talking about threads, it is important for the user to assign the right amount of stack to each defined thread. Especially with limited memory locations owing to the IoT angle, RIOT lets you find your way through this hurdle by introducing you to sample stack sizes that you can use as a base and accordingly customise your board.

A modular and configurable stack for communication

This free software offers support to relevant IoT standards and protocols you would need while coding for your application. It is IPv6 over Low-power Wireless Personal Area Networks (6LoWPAN)—request for comments (RFC) 6282 and RFC6775 compliant, Routing Protocol for

Low-Power and Lossy Networks (RPL)—RFC6550 compliant, and supports OpenWSN, Constrained Application Protocol (CoAP), Concise Binary Object Representation (CBOR) and Universal Binary JSON (UBJSON) standards.

RIOT is also compatible with Arduino API. To enhance capabilities, the OS promotes networking through tap interfaces and offers support to additional tools like Desvirt enabling network virtualisation, RIOT-TV for visualisation and packet capture based wireless sniffer.

Supporting a variety of integration

RIOT can be configured for entire platforms like TelosB or STM32 discovery boards, allows connection to various microcontrollers and a wide range of sensors. Environmental sensors, battery gas gauge, accelerometers, gyroscopes, ultrasonic range finders and servometers are some of the components you will find easy to integrate.

Drivers for interfacing devices can be added as modules to the code, many of which are readily